

SUNTAR, Russian Federation

WMO: 247380

Lat: **62.150N**Lon: **117.650E**Elev: **133**StdP: **99.74**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.6	-47.0	-50.7	0.0	-49.6	-48.4	0.0	-47.0	6.1	-17.1	5.4	-17.9	0.5	20	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.4	19.7	28.0	18.7	25.8	17.7	20.9	27.8	19.8	26.2	18.6	24.2	2.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.6	23.7	17.4	12.6	22.6	16.3	11.8	21.6	61.2	28.0	57.3	26.3	53.6	24.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.6	-50.7	33.7	3.4	1.7	-53.2	35.0	-55.1	36.0	-57.1	36.9	-59.5	38.2		
(5)			-50.7	22.7	3.4	1.2	-53.2	23.6	-55.1	24.3	-57.1	25.0	-59.5	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.4	-31.8	-27.4	-16.6	-3.8	6.6	15.9	18.4	14.7	6.0	-5.7	-22.2	-31.7
	DBStd	19.49	9.18	8.32	8.60	6.09	4.59	4.66	3.79	3.73	4.27	6.87	9.54	9.72
	HDD10.0	6601	1296	1047	824	414	126	7	1	7	133	486	966	1293
	HDD18.3	9098	1554	1281	1083	664	364	101	46	123	369	744	1216	1552
	CDD10.0	629	0	0	0	0	20	183	262	151	13	0	0	0
	CDD18.3	85	0	0	0	0	0	27	49	9	0	0	0	0
	CDH23.3	1086	0	0	0	0	8	369	574	132	2	0	0	0
	CDH26.7	342	0	0	0	0	1	118	194	29	0	0	0	0
Wind	WSAvg	1.9	1.5	1.7	1.7	2.2	2.5	2.1	1.8	1.8	2.1	2.1	1.7	1.5
Precipitation	PrecAvg	285	14	9	8	11	26	39	52	41	31	21	19	16
	PrecMax	457	28	20	21	36	72	110	132	88	116	53	41	34
	PrecMin	176	2	1	0	0	3	10	9	6	4	4	6	5
	PrecStd	61	6	4	5	8	15	19	32	20	19	10	7	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.9	-3.4	6.4	14.5	24.5	32.2	33.4	29.9	22.0	9.2	1.5	-4.3
		MCWB	-8.3	-4.6	1.7	6.8	13.7	19.5	21.1	19.6	14.1	6.0	0.3	-4.5
	2%	DB	-12.6	-8.0	1.9	10.6	21.1	29.9	31.1	26.7	18.3	6.4	-0.8	-9.6
		MCWB	-13.0	-8.7	-1.3	4.5	11.5	18.9	20.3	18.2	12.0	3.7	-1.6	-9.9
	5%	DB	-14.9	-11.4	-1.0	7.8	18.3	27.7	28.7	24.4	15.4	4.4	-3.5	-13.6
		MCWB	-15.2	-12.0	-3.1	2.9	9.8	18.0	19.3	17.1	10.4	2.3	-4.2	-13.8
	10%	DB	-18.3	-14.5	-3.3	5.5	15.6	25.1	26.6	22.1	13.1	2.5	-7.8	-17.4
		MCWB	-18.5	-15.0	-4.9	1.5	8.4	16.7	18.5	16.1	9.3	0.8	-8.4	-17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.3	-4.7	2.1	7.1	14.8	21.9	22.7	20.8	14.7	6.2	0.4	-4.4
		MCDB	-8.0	-3.5	6.7	13.6	22.3	28.5	29.8	26.7	20.5	8.5	1.3	-4.2
	2%	WB	-12.8	-8.8	-1.2	4.8	12.3	20.2	21.4	19.2	12.7	4.1	-1.6	-10.0
		MCDB	-12.6	-8.0	1.4	9.8	19.4	27.9	28.5	24.5	16.8	5.7	-0.9	-9.8
	5%	WB	-15.2	-12.0	-3.1	3.1	10.4	18.8	20.4	18.1	11.1	2.5	-4.3	-13.9
		MCDB	-14.9	-11.5	-1.1	7.3	17.1	25.7	27.1	22.9	14.2	4.2	-3.5	-13.7
	10%	WB	-18.5	-14.9	-5.0	1.7	8.8	17.4	19.4	17.0	9.7	1.0	-8.4	-17.6
		MCDB	-18.3	-14.5	-3.3	5.1	15.0	23.6	25.3	21.0	13.0	2.4	-7.8	-17.4
Mean Daily Temperature Range	5% DB	MDBR	8.1	11.4	15.4	13.0	11.0	11.8	11.4	11.1	9.0	7.3	8.5	7.5
		MCDBR	10.3	11.4	12.6	14.6	16.1	15.6	15.7	16.1	13.3	8.2	8.3	10.0
		MCWBR	10.2	10.8	10.1	9.5	8.4	6.8	6.7	8.0	7.6	6.1	8.1	9.7
		MCDDBR	10.2	11.2	11.9	12.6	14.4	14.0	13.3	13.2	11.5	7.5	8.4	9.8
	5% WB	MCWBR	10.1	10.7	9.6	8.3	8.2	6.9	6.5	7.3	7.1	6.0	8.2	9.7
Clear-Sky Solar Irradiance	taub		0.229	0.267	0.299	0.339	0.373	0.414	0.470	0.429	0.330	0.286	0.226	0.196
	taud		1.939	2.051	2.080	2.134	2.306	2.282	2.100	2.190	2.463	2.352	1.992	1.878
	Ebn at Noon		419	676	796	840	843	810	741	740	780	674	421	266
	Edh at Noon		39	73	104	122	113	118	138	116	73	55	35	22
All-Sky Solar Radiation	RadAvg		0.33	1.30	3.05	4.96	5.33	5.70	5.21	4.04	2.47	1.14	0.47	0.16
	RadStd		0.02	0.06	0.12	0.25	0.33	0.38	0.35	0.25	0.28	0.10	0.05	0.01